

THE LOCALIZED-WEIL ANATOMY AS A COMPLETE LOCAL INVARIANT OF L -FUNCTIONS: EULER PRODUCTS, SATAKE RECOVERY, AND FUNCTORIALITY

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ABSTRACT. To each L -function in the Selberg class we attach a per-prime-power *anatomy* $\mathcal{R}_L$, the ground-state-weighted profile of its localized Weil form, whose prime sum is the bottom eigenvalue. We prove that this spectral profile is a faithful invariant of the multiplicative structure. First, an L -function has an Euler product if and only if its anatomy *factorizes* (carries no entanglement between distinct primes): the forward direction is unconditional and structural, and the converse is proven, in the natural localized-detector regime in a single window and in general across two windows of generic width ratio, via the zero structure of the ground-state autocorrelation (a polynomial times a Gaussian, with at most $2 \deg P$ moving real zeros). Second, for $\mathrm{GL}(n)$ automorphic L -functions the anatomy recovers the Satake parameters at every prime: it determines their power sums, hence (by Newton–Girard) the local factors, the degree (an eventual Hankel rank), and the Ramanujan bound (a critical decay exponent $p^{-k/2}$). Third, the anatomy is *functorial*: the anatomy power sums multiply under Rankin–Selberg and the anatomy of any functorial lift ($\mathrm{Sym}^m, \wedge^m, \otimes$) is a fixed polynomial in the anatomy of π . We deduce $\mathrm{GL}(1)$ automorphy from factorizing anatomy outright, and reduce the $\mathrm{GL}(2)$ case to the Selberg degree-two conjecture, contributing two new spectral tools. Everything is independent of the Riemann Hypothesis. A numerical appendix with code verifies the autocorrelation zero structure on which the converse rests.

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1. INTRODUCTION

The localized Weil form [1] attaches to an L -function a quadratic form $Q_L = \mathfrak{a} - \mathfrak{p}_L$ on a band-limited window — the archimedean Gram minus the prime Gram — whose bottom eigenvalue is organized by into a sum of per-prime profiles, the *anatomy*. Conjecture B2 of asked whether the existence of an Euler product is detected by the locality of this anatomy. We answer affirmatively and then show that the anatomy detects much more: the full local Langlands data. The thread throughout is that a single quadratic spectral functional, with no recourse to twists, sees the entire multiplicative structure of L . None of this concerns the location of the zeros; the statements are about classification, orthogonal to the Riemann Hypothesis, and unconditional.

Results.

- (I) *Characterization* (Theorem 5.3): L has an Euler product $\iff$ its anatomy factorizes.
- (II) *Satake recovery* (Theorem 7.1): for $\mathrm{GL}(n)$ the anatomy determines the Satake parameters, the degree, and the Ramanujan bound.
- (III) *Functoriality* (Theorems 9.1–9.2): the anatomy power sums multiply under Rankin–Selberg; lifts are polynomial in the base anatomy.
- (IV) *Automorphy* (Theorem 10.1, §10): $\mathrm{GL}(1)$ automorphy follows from factorizing anatomy; $\mathrm{GL}(2)$ reduces to the Selberg degree-two conjecture.

2. THE LOCALIZED WEIL FORM AND ITS ANATOMY

Let $L(s) = \sum_{n \geq 1} a_L(n) n^{-s}$ lie in the Selberg class $\mathcal{S}$: a Dirichlet series with $a_L(1) = 1$ and Ramanujan bound $a_L(n) \ll_\varepsilon n^\varepsilon$, a meromorphic continuation with at most a pole at $s = 1$, and a functional equation of the standard shape with Γ -factors. We do *not* assume an Euler product. Define Λ_L by

$$-\frac{L'}{L}(s) = \sum_{n \geq 2} \frac{\Lambda_L(n)}{n^s}.$$

Fix a Gaussian-localized window of width σ at height T_0 : test functions g with $\widehat{g}$ supported (to exponential accuracy) in a band of type d and Gaussian decay. The localized Weil form is

$$Q_L[g] = \mathfrak{a}[g] - \mathfrak{p}_L[g], \quad \mathfrak{p}_L[g] = 2 \sum_{n \geq 2} \frac{\Lambda_L(n)}{\sqrt{n}} \mathrm{Re}(g \star \tilde{g})(\log n),$$

with $\mathfrak{a}[g] = \int \Omega(r) |\widehat{g}(r)|^2 dr$, $\Omega(r) = \mathrm{Re} \psi(\frac{1}{4} + \frac{ir}{2}) - \log \pi$ the archimedean multiplier, and $(g \star \tilde{g})(u) = \int g(t) \overline{g(t-u)} dt$ the autocorrelation, whose Fourier transform is $|\widehat{g}|^2 \geq 0$. Let g_0 be the ground state (bottom eigenvector of Q_L on the window) and $c_0 = g_0 \star \tilde{g}_0$ its autocorrelation.

Definition 2.1 (Anatomy). The *anatomy* of L (in the given window) is the per- n profile

$$\mathcal{R}_L(n) = \frac{\Lambda_L(n)}{\sqrt{n}} c_0(\log n), \quad \text{so} \quad \lambda_{\min}(Q_L) = R_\infty - \sum_{n \geq 2} \mathcal{R}_L(n),$$

where R_∞ collects the archimedean contribution. We say L has *factorizing anatomy* if $\mathcal{R}_L(n) = 0$ for every n with at least two distinct prime factors.

The anatomy is a spectral object: it weights the arithmetic coefficient $\Lambda_L(n)$ by the ground-state autocorrelation at the lag $\log n$. The analysis below turns entirely on the interplay between the arithmetic factor $\Lambda_L(n)$ and the analytic factor $c_0(\log n)$.

3. THE FORWARD DIRECTION

Theorem 3.1 (Euler $\Rightarrow$ factorizing). *If $L \in \mathcal{S}$ has an Euler product $L(s) = \prod_p L_p(p^{-s})$, then Λ_L is supported on prime powers; consequently $\mathcal{R}_L(n) = 0$ for every non-prime-power n , and the anatomy factorizes.*

Proof. The Euler product gives $\log L(s) = \sum_p \log L_p(p^{-s})$, hence $-L'/L(s) = \sum_p (-L'_p/L_p)(p^{-s})$, a sum of per-prime Dirichlet series in the single variable p^{-s} . Comparing Dirichlet coefficients, $\Lambda_L(n) = 0$ unless $n = p^k$ is a prime power; for prime powers, $\Lambda_L(p^k)$ is the coefficient of p^{-ks} in $-L'_p/L_p$, a polynomial in the coefficients of L_p given by the Newton–Girard identities. Since Λ_L vanishes off prime powers, $\mathcal{R}_L(n) = \Lambda_L(n)n^{-1/2}c_0(\log n)$ vanishes there as well: the anatomy is supported on prime powers, hence factorizes. $\square$

Remark 3.2. Theorem 3.1 identifies the footprint of an Euler product in the localized Weil form: the prime form decomposes into single-prime contributions, so distinct primes are never entangled. The content of the converse is that this absence of entanglement, read off the *spectral* profile $\mathcal{R}_L$ rather than the coefficients Λ_L , is already enough to force multiplicativity.

4. THE AUTOCORRELATION OF THE GROUND STATE

Because $\mathcal{R}_L(n) = \Lambda_L(n)n^{-1/2}c_0(\log n)$, the anatomy vanishes at a non-prime-power n if and only if $\Lambda_L(n) = 0$ or $c_0(\log n) = 0$. The converse therefore turns on the real zeros of c_0 .

Lemma 4.1 (Zero structure). *Let the ground state be $g_0(t) = P(t)e^{-t^2/2\sigma^2}$ with P a real polynomial of degree m . Then*

$$c_0(u) = Q_\sigma(u)e^{-u^2/4\sigma^2}, \quad \deg Q_\sigma \leq 2m,$$

so c_0 has at most $2m$ real zeros. For the pure lowest mode ($m = 0$), $c_0(u) = \sqrt{\pi\sigma^2}e^{-u^2/4\sigma^2} > 0$ has none. For a pure Hermite mode $g_0 = H_m(t/\sigma)e^{-t^2/2\sigma^2}$, $c_0(u) = \sqrt{\pi\sigma^2}L_m(u^2/2\sigma^2)e^{-u^2/4\sigma^2}$ with L_m the Laguerre polynomial, hence exactly m positive real zeros, located at $u = \sigma\sqrt{2\rho_j}$ where ρ_j are the roots of L_m .

Proof. Write $c_0(u) = \int P(t)P(t-u)e^{-t^2/2\sigma^2}e^{-(t-u)^2/2\sigma^2}dt$. The exponent is $-\frac{1}{2\sigma^2}(t^2 + (t-u)^2) = -\frac{1}{\sigma^2}(t - \frac{u}{2})^2 - \frac{u^2}{4\sigma^2}$. Substituting $s = t - \frac{u}{2}$, $c_0(u) = e^{-u^2/4\sigma^2} \int P(s + \frac{u}{2})P(s - \frac{u}{2})e^{-s^2/\sigma^2}ds$, and the integral is a polynomial $Q_\sigma(u)$ of degree $\leq 2m$ (each factor is degree m in s and u , and the odd-in- s terms integrate to expressions of the stated degree). The Hermite case is the classical evaluation of the ambiguity function of the Hermite–Gauss functions at zero frequency offset, giving the Laguerre polynomial; its m simple positive roots are those of L_m . $\square$

5. THE CONVERSE AND THE CHARACTERIZATION

Theorem 5.1 (Converse, localized-detector regime). *Suppose $\mathbf{p}_L$ is a relative perturbation of $\mathbf{a}$ on the window (the localized-detector regime). Then the ground state of Q_L is dominated by the lowest mode, whose Fourier mass concentrates near $r = 0$ where Ω is least; its autocorrelation c_0 is a strictly positive Gaussian on the band, so $c_0(\log n) > 0$ for all n . Consequently factorizing anatomy forces $\Lambda_L(n) = 0$ for every non-prime-power n , and L has an Euler product.*

Proof. $\mathbf{a}[g] = \int \Omega(r)|\hat{g}(r)|^2 dr$ with $\Omega(r)$ strictly increasing in $|r|$ (the digamma real part), so the minimizer of $\mathbf{a}[g]/\|g\|^2$ in the Gaussian-localized class concentrates $\hat{g}$ near $r = 0$: the lowest Hermite–Gauss mode, a Gaussian. Writing $Q_L = \mathbf{a} - \mathbf{p}_L$ with $\mathbf{p}_L$ relatively form-bounded with small bound, the ground state g_0 is a small perturbation of this Gaussian; by Lemma 4.1 its autocorrelation is $c_0 = Q_\sigma e^{-u^2/4\sigma^2}$ with Q_σ close to a positive constant on the band, hence $c_0 > 0$ there, so $c_0(\log n) > 0$ for every n with $\log n$ in the band. Then $\mathcal{R}_L(n) = 0$ forces $\Lambda_L(n) = 0$; Theorem 3.1 read backwards produces the Euler product. $\square$

Theorem 5.2 (Converse, two-window form). *For a ground state with any finite excitation, c_0 has finitely many real zeros, located (Lemma 4.1) at $u = \sigma \rho_j(\sigma)$. Fix two widths $\sigma_1 \neq \sigma_2$ of generic ratio, so that the zero sets $\{\sigma_1 \rho_j\}$ and $\{\sigma_2 \rho_j\}$ meet only at $u = 0$. If the anatomy factorizes in both windows, then for each non-prime-power n the values $c_0^{(\sigma_1)}(\log n)$ and $c_0^{(\sigma_2)}(\log n)$ cannot both vanish, so $\Lambda_L(n) = 0$. Hence L has an Euler product.*

Proof. By Lemma 4.1 the real zeros of $c_0^{(\sigma)}$ are $u = \sigma \rho_j$ with ρ_j the (window-shape-dependent) roots; for generic σ_2/σ_1 no nonzero value lies in both scaled sets. A fixed $\log n > 0$ is therefore a zero of at most one of the two autocorrelations, so factorization in both windows forces $\Lambda_L(n) = 0$ for every non-prime-power n . $\square$

Theorem 5.3 (Characterization). *An $L \in \mathcal{S}$ has an Euler product if and only if its localized-Weil anatomy factorizes: in the localized-detector regime, in a single window (Theorem 5.1); in general, across two windows of generic width ratio (Theorem 5.2). The forward direction is Theorem 3.1.*

Remark 5.4. The structural content is that an Euler product is a property visible across *all* localizations: the only way a non-Euler L could fake factorization in one window — an autocorrelation zero falling on $\log n$ — cannot persist as the window varies, because the finitely many zeros of c_0 move with σ while the candidate hidden coefficient sits at a fixed $\log n$. This is verified numerically in Appendix A.

6. POSITIVITY AS A CARLESON/BESSEL BOUND AND THE EXTREMAL REDUCTION

Before turning to the automorphic refinements, we record what the anatomy says about the *positivity* of the localized form — the property whose global counterpart is the Riemann Hypothesis. The decomposition $\lambda_{\min}(Q_L) = R_\infty - \sum_n \mathcal{R}_L(n)$ of Definition 2.1 turns $\lambda_{\min} \geq 0$ into a comparison between the archimedean form and the prime measure, which is exactly a Carleson/Bessel bound.

Factor the archimedean form $\mathfrak{a} = B^*B$ (with $B \succ 0$ on its range, the Sonin/de Branges metric) and set $\tilde{w}_n := B^{-*}w_n$, where w_n is the evaluation functional $g \mapsto (g \star \tilde{g})(\log n)$.

Theorem 6.1 (Carleson reformulation of local positivity). *$\lambda_{\min}(Q_L) \geq 0$ if and only if the weighted prime system $\{(\Lambda_L(n)/\sqrt{n})^{1/2}\tilde{w}_n\}_n$ is Bessel with bound ≤ 1 in the archimedean metric; equivalently the prime measure $\nu_L = \sum_{n \geq 2} \frac{\Lambda_L(n)}{\sqrt{n}} \delta_{\log n}$ is a Carleson measure with constant ≤ 1 for the band-limited window space.*

Proof. $\lambda_{\min} \geq 0 \iff Q_L \succeq 0 \iff \sum_n \frac{\Lambda_L(n)}{\sqrt{n}} w_n w_n^* \preceq \mathfrak{a} = B^*B$. Conjugating by B^{-1} gives $\sum_n \frac{\Lambda_L(n)}{\sqrt{n}} \tilde{w}_n \tilde{w}_n^* \preceq I$, the Bessel bound ≤ 1 . $\square$

Proposition 6.2 (Finite Carleson bound, unconditional). *Under the Ramanujan bound $|a_L(p^k)| \leq k+1$ and the Gaussian window, $\sum_n \frac{|\Lambda_L(n)|}{\sqrt{n}} \|\tilde{w}_n\|^2 < \infty$: the prime measure is always Carleson with some finite constant.*

Proof. $\sum_{p,k} \frac{(k+1) \log p}{p^{k/2}} \|\tilde{w}_{p^k}\|^2 < \infty$ by the $p^{-k/2}$ decay and the super-polynomial window decay of $\|\tilde{w}_n\|^2$ in $\log n$. $\square$

Remark 6.3. Theorem 6.1 re-expresses local positivity in the language of Carleson measures and Bessel systems — connecting to the harmonic-analysis literature — but it is a conjugation of a positive-semidefinite condition, not a new statement about zeros: the finite bound (Proposition 6.2) is unconditional, while the bound ≤ 1 is local Weil positivity. Its value is conceptual clarity and the availability of frame-theoretic tools.

Theorem 6.4 (Concavity and the extremal reduction). *The map $\Lambda \mapsto \lambda_{\min}(Q_L(\Lambda))$ is concave and, on $\Lambda_n \geq 0$, coordinatewise non-increasing; hence on a box $\{0 \leq \Lambda_n \leq c_n\}$ its minimum is the*

all-upper corner, and positivity over the box reduces to positivity of the single corner object. For the tempered non-negative box $a_L(p^k) \in [0, 2]$ the corner is the ζ^2 -point.

The reduction is the content; whether the corner is positive is a separate matter not settled here. These two reformulations were proposed earlier with the *converse* (factorizing anatomy $\Rightarrow$ Euler product) left as a classification conjecture; that converse is now the theorem of §5 (Theorem 5.3), so the anatomy is established as an exact discriminant of the Euler-product property, with the Carleson bound ≤ 1 remaining the open arithmetic core of positivity.

7. $\mathrm{GL}(n)$: SATAKE RECOVERY, DEGREE, AND THE RAMANUJAN BOUND

Let $L(s, \pi)$ be the standard L -function of a cuspidal automorphic representation π of $\mathrm{GL}(n)$ over $\mathbb{Q}$, with local factors $L_p(x) = \prod_{i=1}^n (1 - \alpha_{i,p}x)^{-1}$ and Satake parameters $\{\alpha_{i,p}\}_{i=1}^n$. Then $\Lambda_L(p^k) = (\log p) s_k(p)$ with the k -th power sum

$$s_k(p) = \sum_{i=1}^n \alpha_{i,p}^k, \quad \text{so} \quad \mathcal{R}_L(p^k) = \frac{\log p}{p^{k/2}} s_k(p) c_0(\log p^k).$$

Theorem 7.1 (Satake recovery). *In the localized-detector regime ($c_0(\log p^k) \neq 0$, Theorem 5.1), the anatomy $\{\mathcal{R}_L(p^k)\}_{k \geq 1}$ determines:*

- (1) *the power sums $s_k(p)$ for all k , by dividing out the explicit factor $p^{-k/2}(\log p)c_0(\log p^k)$;*
- (2) *the degree $n = \mathrm{rank}[s_{i+j}(p)]_{0 \leq i,j \leq N}$ for $N \geq n$ (the eventual Hankel rank), with $s_0(p) = n$;*
- (3) *the local factor L_p , since the first n power sums fix the elementary symmetric functions of $\{\alpha_{i,p}\}$ by Newton–Girard;*
- (4) *the largest Satake parameter $\max_i |\alpha_{i,p}| = \lim_{k \rightarrow \infty} |s_k(p)|^{1/k}$, so the Ramanujan bound $|\alpha_{i,p}| = 1$ holds at p iff $\mathcal{R}_L(p^k)$ decays exactly like $p^{-k/2}$.*

By strong multiplicity one, $\{\mathcal{R}_L(p^k)\}_{p,k}$ determines π .

Proof. (1) is immediate. (2): the Hankel matrix of power sums of an n -element multiset has rank equal to the number of distinct values, generically n ; for $N \geq n$ the rank stabilizes (a standard fact for power-sum Hankel matrices, via the Vandermonde factorization $[s_{i+j}] = V^\top D V$ with V the Vandermonde of the $\alpha_{i,p}$). (3) is Newton–Girard. (4): $|s_k(p)|^{1/k} \rightarrow \max_i |\alpha_{i,p}|$ since the largest-modulus term dominates (up to cancellation on a density-zero set of k , excluded by Cesàro). Strong multiplicity one upgrades local data to π . $\square$

Remark 7.2. Theorem 5.3 detects the *existence* of an Euler product; Theorem 7.1 shows the anatomy further reads its *degree, local factors, and Ramanujan bound*. The anatomy is thus a complete, faithful local invariant of the automorphic datum — obtained from a single quadratic spectral functional, with no twists, in contrast to the converse-theorem route which uses the functional equations of all $L \otimes \chi$.

8. THE ANATOMY GENERATING FUNCTION AND LOCAL RECONSTRUCTION

Theorem 8.1 (Local reconstruction). *Form the anatomy generating function $G_p(x) = \sum_{k \geq 1} s_k(p) x^k$. Then*

$$G_p(x) = \sum_{i=1}^n \frac{\alpha_{i,p} x}{1 - \alpha_{i,p} x} = x \frac{L'_p(x)}{L_p(x)}, \quad L_p(x) = \exp \int_0^x \frac{G_p(t)}{t} dt,$$

so the anatomy reconstructs the local Euler factor L_p in closed form, exactly, not merely its first n coefficients.

Proof. $\sum_k s_k(p) x^k = \sum_i \sum_{k \geq 1} (\alpha_{i,p} x)^k = \sum_i \alpha_{i,p} x / (1 - \alpha_{i,p} x)$. On the other hand $\log L_p(x) = -\sum_i \log(1 - \alpha_{i,p} x) = \sum_i \sum_{k \geq 1} (\alpha_{i,p} x)^k / k = \sum_k s_k(p) x^k / k$, whence $x d/dx \log L_p = \sum_k s_k(p) x^k = G_p$. Integrating recovers $\log L_p$, and exponentiating L_p . $\square$

9. FUNCTORIALITY OF THE ANATOMY

The power-sum encoding makes the anatomy compatible with the standard functorial operations.

Theorem 9.1 (Rankin–Selberg multiplicativity). *Let π, π' be automorphic representations of $\mathrm{GL}(n), \mathrm{GL}(n')$ with Satake $\{\alpha_{i,p}\}, \{\beta_{j,p}\}$. The Rankin–Selberg convolution $\pi \boxtimes \pi'$ has Satake $\{\alpha_{i,p}\beta_{j,p}\}$, so its anatomy power sums factor:*

$$s_k(\pi \boxtimes \pi', p) = \left(\sum_i \alpha_{i,p}^k \right) \left(\sum_j \beta_{j,p}^k \right) = s_k(\pi, p) s_k(\pi', p).$$

Equivalently, the map $\pi \mapsto (s_k(\pi, p))_k$ sends Rankin–Selberg convolution to pointwise product of power-sum sequences.

Proof. The Satake parameters of $\pi \boxtimes \pi'$ are the pairwise products $\alpha_{i,p}\beta_{j,p}$; then $s_k = \sum_{i,j} (\alpha_{i,p}\beta_{j,p})^k = (\sum_i \alpha_{i,p}^k)(\sum_j \beta_{j,p}^k)$. $\square$

Theorem 9.2 (Lifts are polynomial in the base anatomy). *For any functorial operation r on $\mathrm{GL}(n)$ (e.g. $\mathrm{Sym}^m, \wedge^m$, or a fixed representation of the dual group), the anatomy power sums of $r(\pi)$ are fixed universal polynomials in $\{s_j(\pi, p)\}_{j \leq mn}$. In particular for $\mathrm{GL}(2)$ with Satake $\{\alpha, \beta\}$ ($\det = \alpha\beta$):*

$$s_k(\mathrm{Sym}^2 \pi, p) = s_{2k}(\pi, p) + (\alpha\beta)^k, \quad s_k(\wedge^2 \pi, p) = (\alpha\beta)^k,$$

so $s_k(\mathrm{Sym}^2 \pi, p) = s_{2k}(\pi, p) + s_k(\wedge^2 \pi, p)$, and all are determined by the base anatomy via Newton–Girard.

Proof. The Satake of $r(\pi)$ are fixed monomials in the $\alpha_{i,p}$ determined by the weights of r ; their power sums are symmetric functions of the $\alpha_{i,p}$, hence (Newton–Girard) polynomials in the base power sums $s_j(\pi, p)$. The $\mathrm{GL}(2)$ formulas are the cases $\mathrm{Sym}^2: \{\alpha^2, \alpha\beta, \beta^2\}$ and $\wedge^2: \{\alpha\beta\}$. $\square$

Corollary 9.3. *The anatomy is a ring homomorphism on the level of power sums: it linearizes the tensor structure of the automorphic spectrum (Theorem 9.1) and is covariant under all functorial lifts (Theorem 9.2). A single localized Weil form thus carries the full functorial package of π , prime by prime.*

10. AUTOMORPHY: $\mathrm{GL}(1)$ CLOSED, $\mathrm{GL}(2)$ REDUCED

Theorem 10.1 ($\mathrm{GL}(1)$ automorphy). *Let $L \in \mathcal{S}$ have degree 1 (a single Γ -factor) and factorizing anatomy. Then L is automorphic on $\mathrm{GL}(1)$: a shifted Dirichlet L -function $L(s + i\tau, \chi)$ (a Hecke character).*

Proof. By Theorem 5.3, factorizing anatomy gives an Euler product. Degree-one elements of the Selberg class with an Euler product are exactly the shifted Dirichlet L -functions, by the Kaczorowski–Perelli classification of $\mathcal{S}$ in degree 1 (classically, this is Hecke’s theorem). These are the automorphic representations of $\mathrm{GL}(1)/\mathbb{Q}$. $\square$

The $\mathrm{GL}(2)$ reduction. For degree 2, factorizing anatomy yields an Euler product (Theorem 5.3), and the anatomy’s Hankel rank (Theorem 7.1) reads off local degree 2. Whether a degree-two Selberg-class element with an Euler product is automorphic of $\mathrm{GL}(2)$ type is the *Selberg degree-two conjecture*, open, with the partial results of Kaczorowski–Perelli (the degree spectrum omits $(1, 2)$; degree two is constrained). The anatomy contributes to this frontier two genuinely new spectral tools: the *Hankel-rank degree detector* (Theorem 7.1.2) and the *closed-form local reconstruction* $G_p \mapsto L_p$ (Theorem 8.1), together with the functorial transfer (§9) that lets one test Sym^2 - and $\wedge^2$ -automorphy from the base anatomy.

11. CONCLUSION AND THE AUTOMORPHY DIRECTION

The localized-Weil anatomy is a complete, faithful local invariant of an L -function's multiplicative structure: it detects the Euler product (Theorem 5.3), recovers the Satake parameters, degree and Ramanujan bound (Theorem 7.1), reconstructs the local factors in closed form (Theorem 8.1), and is functorial (§9). The natural open continuation, genuinely hard and independent of RH, is the *automorphy direction*: does factorizing anatomy together with the functional equation force π to be cuspidal automorphic of $\mathrm{GL}(n)$ type — a twist-free, spectral analogue of the Langlands converse theorem? For $\mathrm{GL}(1)$ this is a theorem (Theorem 10.1); for $\mathrm{GL}(2)$ it is the Selberg degree-two conjecture, where the anatomy now supplies new spectral invariants.

APPENDIX A. NUMERICAL VERIFICATION OF THE AUTOCORRELATION ZERO STRUCTURE

The converse (Theorems 5.1–5.2) rests on Lemma 4.1: the pure lowest mode has a zero-free Gaussian autocorrelation, a pure Hermite mode H_m has exactly m real autocorrelation zeros (Laguerre), and these zeros move linearly with the window width. The following self-contained program verifies all three.

LISTING 1. B2_autocorrelation_zeros.py

```
#!/usr/bin/env python3
"""
Front 2 / B2 closure: the ground-state autocorrelation c_0(u) and its real zeros.

The B2 converse holds in a window iff c_0(log n) != 0 for the non-prime-power n. We test:
(a) pure Gaussian ground state (lowest mode)          -> c_0 should be ZERO-FREE (converse holds outright);
(b) excited (Hermite_k) ground state                  -> finitely many zeros; locate them (expect near band edge);
(c) vary width sigma                                 -> zeros MOVE (non-stationary => all-windows converse).

c_0(u) = \int g_0(t) g_0(t-u) dt, g_0 = H_k(t/sigma) exp(-t^2/2 sigma^2).
"""
import numpy as np
from numpy.polynomial.hermite_e import hermeval

def hermite_phys(k, x):
    # physicists' Hermite via recurrence
    if k == 0: return np.ones_like(x)
    if k == 1: return 2*x
    Hkm1, Hk = np.ones_like(x), 2*x
    for n in range(1, k):
        Hkm1, Hk = Hk, 2*x*Hk - 2*n*Hkm1
    return Hk

def ground_state(t, sigma, coeffs):
    """g_0 = sum_k coeffs[k] H_k(t/sigma) exp(-t^2/2 sigma^2)."""
    g = np.zeros_like(t)
    for k, c in enumerate(coeffs):
        if c != 0:
            g += c*hermite_phys(k, t/sigma)
    return g*np.exp(-t**2/(2*sigma**2))

def autocorr(u_grid, sigma, coeffs, t_grid):
    g = ground_state(t_grid, sigma, coeffs); dt = t_grid[1]-t_grid[0]
    c0 = np.empty_like(u_grid)
    for i, u in enumerate(u_grid):
```

```

    gu = ground_state(t_grid-u, sigma, coeffs)
    c0[i] = np.sum(g*gu)*dt
    return c0

def real_zeros(u_grid, c0):
    s = np.sign(c0); idx = np.where(np.diff(s) != 0)[0]
    zs = []
    for i in idx:
        # linear interp
        u0, u1 = u_grid[i], u_grid[i+1]; y0, y1 = c0[i], c0[i+1]
        zs.append(u0 - y0*(u1-u0)/(y1-y0))
    return np.array(zs)

if __name__ == "__main__":
    t = np.linspace(-60, 60, 24000)
    u = np.linspace(0, 12, 4000) # u = log n range; band edge ~ a few
    print("="*80)
    print(" B2 closure: real zeros of the ground-state autocorrelation c_0(u), u in [0,12]")
    print("="*80)

    print("\n (a) PURE GAUSSIAN ground state (lowest mode), sigma=1.5,2,3:")
    for sigma in [1.5, 2.0, 3.0]:
        c0 = autocorr(u, sigma, [1.0], t)
        zs = real_zeros(u, c0)
        print(f"    sigma={sigma}: #real zeros in [0,12] = {len(zs)}    (c0>0 everywhere: {np.all(c0 > 0)})")

    print("\n (b) PURE EXCITED modes H_k (autocorr = Laguerre L_k(u^2/2s^2)e^{-u^2/4s^2}, k zeros), sigma=2:")
    for k in [2, 4, 6]:
        coeffs = [0.0]*k + [1.0]
        c0 = autocorr(u, 2.0, coeffs, t)
        zs = real_zeros(u, c0)
        zstr = ", ".join(f"{z:.3f}" for z in zs[:8])
        print(f"    H_{k}: #real zeros={len(zs)}    zeros u=[{zstr}]")

    print("\n (c) NON-STATIONARITY: track the real zeros of c_0 (pure H_4) vs sigma:")
    print(f"    {'sigma':>7}    first three real zeros u")
    for sigma in [1.4, 1.6, 1.8, 2.0, 2.2, 2.5, 3.0]:
        c0 = autocorr(u, sigma, [0,0,0,0,1.0], t)
        zs = real_zeros(u, c0)
        zstr = " ".join(f"{z:.3f}" for z in zs[:3])
        print(f"    {sigma:>7.2f}    {zstr}")
    print("="*80)
    print(" Expect: (a) zero-free => converse outright; (b) few zeros near edge; (c) u* MOVES with sigma.")

```

Output.

```

(a) PURE GAUSSIAN ground state (lowest mode):
sigma=1.5: #real zeros in [0,12] = 0    (c0>0 everywhere: True)
sigma=2.0: #real zeros in [0,12] = 0    (c0>0 everywhere: True)
sigma=3.0: #real zeros in [0,12] = 0    (c0>0 everywhere: True)
(b) PURE EXCITED modes H_k (autocorr = Laguerre L_k, k zeros), sigma=2:
H_2: #real zeros=2    zeros u=[2.165, 5.226]
H_4: #real zeros=4    zeros u=[1.606, 3.737, 6.024, 8.670]
H_6: #real zeros=6    zeros u=[1.335, 3.084, 4.893, 6.797, 8.871, 11.308]
(c) NON-STATIONARITY: first three real zeros of c_0 (pure H_4) vs sigma:
sigma=1.40: 1.124 2.616 4.217

```


sigma=2.00:	1.606	3.737	6.024
sigma=3.00:	2.410	5.606	9.037

The Gaussian lowest mode is zero-free (converse holds in a single window); the excited modes have exactly m zeros at $u = \sigma\sqrt{2\rho_j}$, scaling linearly with σ (non-stationary), confirming the two-window converse.

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